CLINICAL · APPROACH TO

Approach to the hoarse voice

Hoarseness, or dysphonia, is any change in the quality of the voice: rough, breathy, strained, weak or tiring. Most of it is inflammatory, functional or due to benign lesions, but persistent hoarseness is the commonest presenting symptom of laryngeal cancer, and a hoarse voice with a paralysed vocal fold may be the first sign of a tumour in the chest or neck. The rule is simple: any adult with hoarseness lasting more than three weeks should have the larynx examined, and the examination should be good enough to see the vocal folds vibrate.

AT A GLANCE

- Over 3 weeks: laryngoscopy.
- Smoker, pain, otalgia, dysphagia, neck node: urgent suspected-cancer pathway.
- Breathy voice and weak cough: palsy — image skull base to arch.
- Normal larynx: think muscle tension dysphonia, but look again if it does not improve.

1 CHAPTER ONE The voice and its failure

How the folds make sound.

1.1 Physiology in brief

The vocal fold has a layered structure. The **cover** (epithelium and the jelly-like superficial lamina propria, Reinke's space) slides over the **transition** (the vocal ligament) and the **body** (the vocalis muscle). Air from the lungs sets the cover vibrating as a *mucosal wave*. The voice fails when the folds cannot close (paralysis, atrophy, a gap), when the cover is stiff or heavy (scar, sulcus, Reinke's oedema, cancer), when a lesion interrupts closure (nodules, polyps, cysts), or when the muscles around the larynx work against it (muscle tension dysphonia, dystonia).

RED FLAGS IN HOARSENESS

- **Hoarseness over three weeks in a smoker or drinker,** or any persistent hoarseness over 45: urgent laryngoscopy on a suspected cancer pathway.
- **Stridor or breathlessness:** airway assessment first.

- **Hoarseness with dysphagia, odynophagia, referred otalgia, haemoptysis, weight loss or a neck lump.**
- **A breathy voice and weak cough after thyroid, chest, neck or cardiac surgery, or with no obvious cause:** vocal fold paralysis — image the vagus and recurrent laryngeal nerve course.
- **Hoarseness in a child with breathing difficulty:** recurrent respiratory papillomatosis.

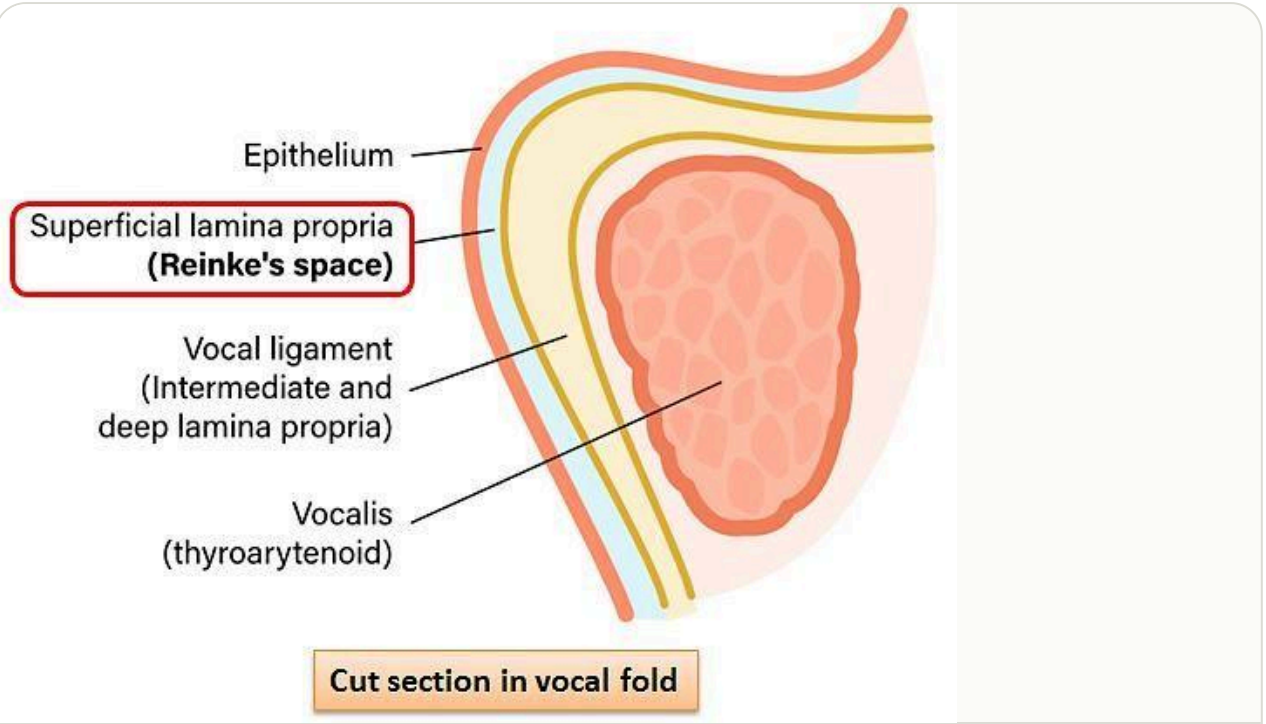


FIGURE 1 Layers of the vocal fold: the cover (epithelium and Reinke's space) over the vocal ligament and muscle.

Anatomy atlas · ENT Steps p. 246

2 CHAPTER TWO History

Onset, use and risk.

2.1 The questions

Ask how the voice changed and when: sudden onset after shouting suggests a haemorrhage or polyp; onset after a cold suggests laryngitis; onset after an operation or intubation suggests nerve injury or arytenoid dislocation; gradual

progression suggests a lesion or cancer. Ask whether it fluctuates, worsens with use or improves with rest. Ask about voice demands (teachers, singers, call-centre workers), smoking and alcohol, reflux symptoms, inhaled steroids, thyroid disease, neurological symptoms (tremor, weakness, swallowing problems), breathing, swallowing, pain and weight. The **Voice Handicap Index** (VHI-10) measures the effect on the patient's life.

3 CHAPTER THREE Examination

See the folds move.

3.1 Laryngeal examination

Listen to the voice: breathy (a gap), rough (irregular vibration), strained (muscle tension or dystonia), tremulous or with voice breaks. Examine the neck for lymph nodes, the thyroid and laryngeal tenderness. Examine the larynx with a **flexible nasendoscope**, ideally a distal-chip scope, looking at the mucosa, any lesion, the movement of each fold and arytenoid, and the pattern of closure; ask the patient to sniff, say "ee" and cough. **Videostroboscopy** shows the mucosal wave and is needed to distinguish nodules from cysts, to detect scar and sulcus, and to judge early cancer. Perceptual rating (GRBAS) and acoustic analysis document the voice.



FIGURE 2 Flexible laryngoscopy in clinic.
Dhingra, fig. 62.6

CHAPTER FOUR

4 Causes

By mechanism.

CAUSES OF HOARSENESS

GROUP	CAUSES	CLUES
Inflammatory	Acute laryngitis, chronic laryngitis, laryngopharyngeal reflux, inhaled steroid laryngitis	Infection, smoking, reflux, inhalers
Benign lesions	Nodules, polyps, cysts, Reinke's oedema, sulcus, granuloma	Voice use, smoking (Reinke's), intubation or reflux (granuloma)
Premalignant and malignant	Leukoplakia, dysplasia, squamous cell carcinoma	Smoker, progressive, white or irregular lesion
Papillomatosis	Recurrent respiratory papillomatosis (HPV 6 and 11)	Children and young adults, recurrence

GROUP	CAUSES	CLUES
Neurological	Vocal fold paralysis, laryngeal dystonia, essential tremor, Parkinson's disease, motor neurone disease, myasthenia	Breathy voice, strangled voice breaks, tremor, other signs
Functional	Muscle tension dysphonia, functional aphonia	Normal folds, strained voice, normal cough
Other	Hypothyroidism, ageing (presbyphonia), arthritis of the cricoarytenoid joint, trauma, amyloid	Context

4.1 Some key causes

Vocal fold nodules are bilateral, symmetrical swellings at the junction of the anterior and middle thirds, caused by voice overuse; voice therapy is the treatment. **Polyps** are usually unilateral, often after a haemorrhage, and are removed by microlaryngoscopy if therapy does not help. **Reinke's oedema** is diffuse swelling of the superficial lamina propria in smokers, giving a low-pitched voice, especially in women; stopping smoking is essential, and surgery reduces the swelling while preserving the cover. **Leukoplakia** carries a risk of malignant change that depends on the grade of dysplasia and is excised for histology.

Vocal fold paralysis is more often left-sided, because the left recurrent laryngeal nerve has a longer course around the aortic arch. Causes are surgery (thyroid, cardiothoracic, anterior cervical spine), malignancy (lung, oesophagus, thyroid, skull base), and idiopathic. Unexplained paralysis is investigated with CT from the skull base to the aortic arch. **Muscle tension dysphonia** is the commonest functional voice disorder, with a strained voice, laryngeal tension and normal or compensatory supraglottic squeezing; it responds to voice therapy. **Laryngeal (spasmodic) dystonia** produces strangled voice breaks, and is treated with botulinum toxin.

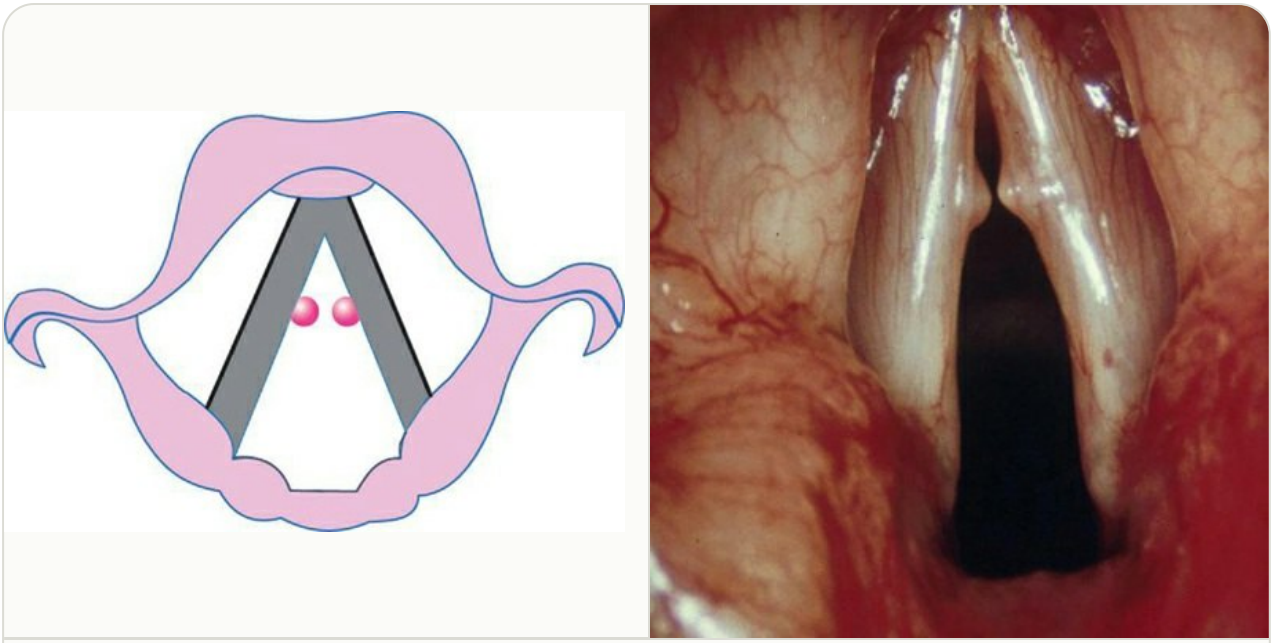


FIGURE 3 First, the site of vocal nodules at the junction of the anterior and middle thirds. Second, nodules on both folds.

Dhingra, figs 61.1 and 61.2

5 CHAPTER FIVE Management

Principles.

5.1 For all patients

Voice hygiene and **voice therapy** with a speech and language therapist are central to most benign and functional voice problems. Smoking cessation, hydration, treatment of reflux when its features are present, and review of inhaled steroid technique are also important.

5.2 Surgery

Phonosurgery under general anaesthesia uses cold steel or lasers to remove lesions while preserving the superficial lamina propria. Suspicious lesions are biopsied or excised for histology. Vocal fold paralysis that does not recover is treated with **injection medialisation** (temporary or long-lasting), **medialisation thyroplasty**, arytenoid adduction or laryngeal reinnervation. Cancer is staged and treated on the head and neck pathway.

THE HOARSE VOICE

1 Red flags or over three weeks?

→ **yes**
laryngoscopy, urgently if smoker, over 45, stridor or other head and neck symptoms

→ **recent cold, improving**
wait and review

2 Laryngoscopy

→ **lesion suspicious for cancer or dysplasia**
microlaryngoscopy and biopsy, staging

→ **benign lesion**
voice therapy, phonosurgery if needed

→ **inflammation**
smoking, reflux and inhaler review

3 Mobility

→ **fold paralysed**
CT skull base to aortic arch, injection or thyroplasty if no recovery

→ **normal folds, strained voice**
muscle tension dysphonia — voice therapy

→ **voice breaks**
dystonia — botulinum toxin

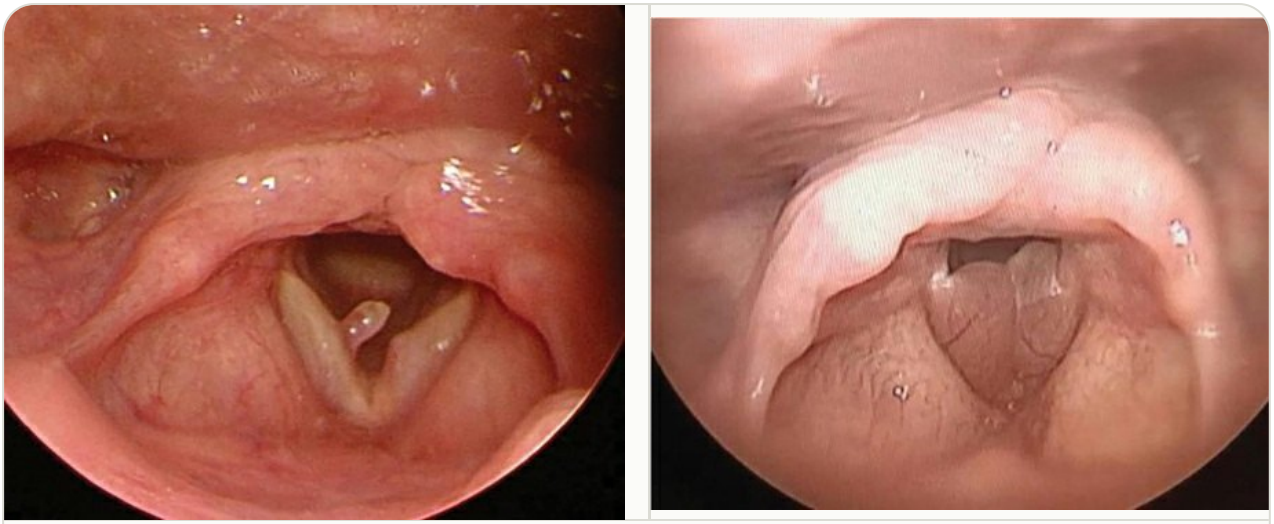


FIGURE 4 First, a pedunculated vocal fold polyp. Second, bilateral Reinke's oedema in a smoker. Dhingra, figs 61.4 and 61.5

CHAPTER SIX

6 For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The three-week rule** and urgent referral for persistent hoarseness.
- **The body–cover model** and the mucosal wave; stroboscopy.
- **Nodules, polyps, cysts and Reinke's oedema:** features and treatment.
- **Vocal fold paralysis:** causes, investigation and treatment options.
- **Functional dysphonia** and the role of voice therapy.

6.2 Common pitfalls

COMMON PITFALLS

- Prescribing antibiotics or a proton pump inhibitor for months without examining the larynx.
- Missing a lung cancer in a patient with left vocal fold paralysis.

- Operating on nodules before voice therapy.
- Stripping the vocal fold and leaving it scarred.

Written for this app from Oxford Textbook of Otolaryngology ch. 47, 79, Oxford Specialist Handbook ch. 3, Oxford Handbook of ENT ch. 3, Dhingra ch. 60, 62. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

LARYNX & TRACHEA · INFECTION & INFLAMMATION

Paralysis of the Recurrent Laryngeal Nerve

IMAGING FINDINGS

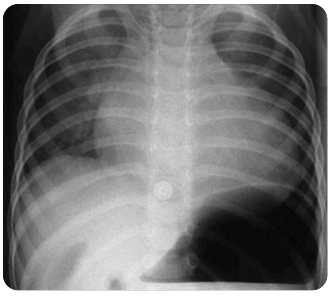
- Medially rotated arytenoid and dilated ventricle
- Dilated pyriform sinus; medial, thickened aryepiglottic fold
- Atrophy of the posterior cricoarytenoid and thyroarytenoid

TOP DIFFERENTIALS

- Arytenoid dislocation after trauma
- Glottic carcinoma
- Laryngocele

CHECKLIST

- Scan from skull base to the aortopulmonary window on the left
- Thyroid, oesophagus and mediastinal nodes



A 7-year-old child with hoarseness and left vocal cord paralysis. Chest radiograph, posteroanterior view. The grossly dilated heart is compressing…



CECT in a patient with new-onset hoarseness shows slightly medially rotated left arytenoid cartilage & a more prominent dilated left laryngeal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 296.

LARYNX & TRACHEA · TUMOURS

Glottic Tumor

IMAGING FINDINGS

- Thickening or mass of the true cord
- Anterior commissure thicker than 1 mm
- Paraglottic and subglottic spread

TOP DIFFERENTIALS

- Benign cord lesion
- Chondrosarcoma (cricoid, chondroid matrix)
- Cord paralysis (medial arytenoid)

CHECKLIST

- Cord movement on breath-hold images
- Thyroid cartilage erosion: inner cortex T3, through it T4a



CT with intravenous contrast enhancement, axial. Slight asymmetry of the vocal cords can be observed with slight enlargement of the vocal…



Axial CECT shows an enhancing right TVC exophytic mass. Anterior and posterior commissures are normal. Right arytenoid cartilage sclerosis is…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 530.

LARYNX & TRACHEA · INFECTION & INFLAMMATION

Laryngocele

IMAGING FINDINGS

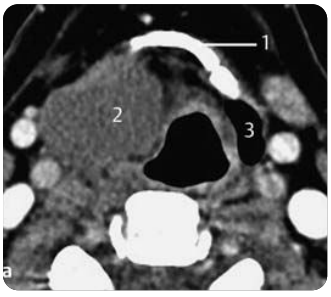
- Air or fluid-filled sac from the laryngeal ventricle
- Internal (paraglottic) or external through the thyrohyoid membrane
- Rim enhancement if infected (pyolaryngocele)

TOP DIFFERENTIALS

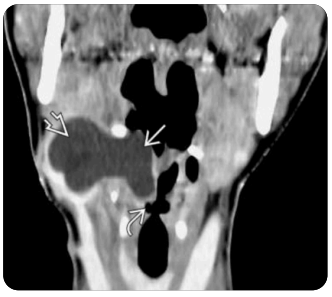
- Thyroglossal duct cyst
- Second branchial cleft cyst
- Saccular cyst

CHECKLIST

- Exclude a carcinoma at the ventricle in adults
- Airway compromise



CT with intravenous contrast enhancement, axial. The axial slice at the lower border of the hyoid (1) reveals a homogeneous, expanding…



Coronal CECT reformat shows a similar lesion. This is also known as a "mixed" laryngocele because it contains internal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 292.